

# Practice 2 Partial 3: Estimation Analysis: Mean and Proportions

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## General Objective

Learn the concept of **statistical estimation** and apply its types (**point estimation** and **interval estimation**) for data analysis in RStudio.

## Practice Development

### Scenario:

1. Simular una muestra de 100 datos, con una desviación estándar de 3.
2. Calcular la proporción de empleados que trabajan menos de 40 horas.
3. Contestar nuevamente las preguntas con los resultados actualizados.
4. Agregar la conclusión nueva según los resultados.
5. Puntos extra: Agregar algún gráfico para representar mejor los datos / proporciones.

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## Step 1: Data Simulation

```
# Generating a random sample of 100 employees
```

```
set.seed(123) # Seed for reproducibility
```

```
work_hours <- rnorm(100, mean = 40, sd = 3)
```

```
# Display first observations
```

```
head(work_hours)
```

```
## [1] 38.31857 39.30947 44.67612 40.21153 40.38786 45.14519
```

## Step 2: Point and Interval Estimation for Proportion

Now estimate the proportion of employees who work less than 40 hours per week:

```
# Number of employees working more than 45 hours
```

```
working_less_40 <- sum(work_hours < 40)
```

```
n <- length(work_hours)
```

```

# Point estimation of the proportion
sample_proportion <- working_less_40 / n
cat("The sample proportion of people working less of 40 hours is: ",
sample_proportion*100, "%")

## The sample proportion of people working less of 40 hours is: 48 %

# Confidence interval for the proportion
prop.test(working_less_40, n, conf.level = 0.95)

##
## 1-sample proportions test with continuity correction
##
## data: working_less_40 out of n, null probability 0.5
## X-squared = 0.09, df = 1, p-value = 0.7642
## alternative hypothesis: true p is not equal to 0.5
## 95 percent confidence interval:
## 0.3798722 0.5816817
## sample estimates:
## p
## 0.48

```

### Step 3: Estimations of the Mean

Calculate the Work\_hours mean:

```

mean <- mean(work_hours)
cat("The point estimation of the mean is: ", mean)

## The point estimation of the mean is: 40.27122

t.test(work_hours, conf.level = 0.95)

##
## One Sample t-test
##
## data: work_hours
## t = 147.06, df = 99, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
## 39.72785 40.81459
## sample estimates:
## mean of x
## 40.27122

```

### Step 4. Answer the following questions based on your results:

- a) What was the sample mean obtained? 40.27122
- b) What was the confidence interval for the mean? 95%

- c) What was the sample proportion of employees working less than 40 hours?  
approximately 48% of employees
- d) What was the confidence interval for the proportion? 95%
- e) Why is the confidence level important in these results? In order to have a certainty for the real mean to be within the proof results

## Step 5. Conclusion

The sample shows that almost half of the employees work less than 40 hours. However, because of the sample, we can't assure that is the real percentage of people working that time. With a certainty of 95%, we can say that the real mean is between 37% and 58% approximately. This indicates that our estimate has considerable variability and it would be advisable to collect more data for great accuracy.

### Extra point: Plot

```
hist(work_hours, freq = TRUE, include.lowest = TRUE, right = TRUE,  
denstiy = NULL, col = "#aaaaff", border = "#000000", main =  
paste("Histogram of working hours") )
```

